

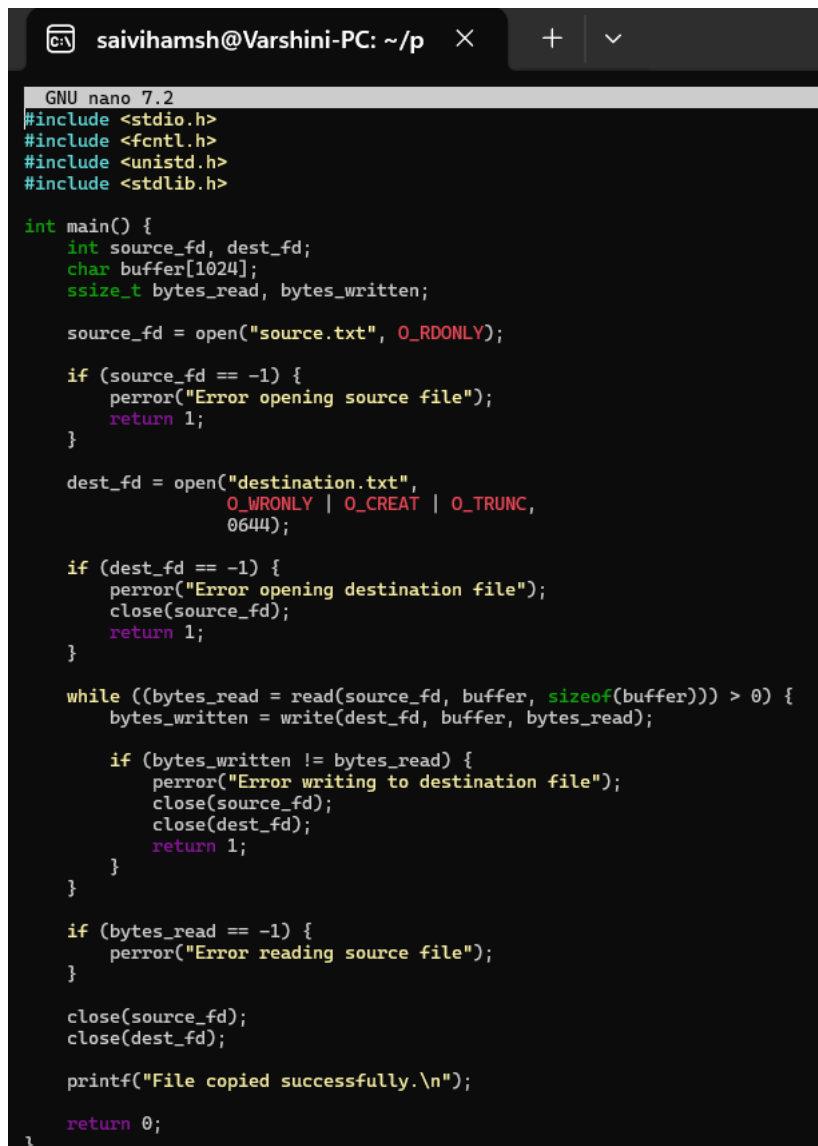
PRACTICAL WEEK-2

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Session-2

Task1: Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

A screenshot of a terminal window with a dark background. The window title bar shows 'saivihamsh@Varshini-PC: ~/p' and standard window controls. The terminal displays the GNU nano 7.2 editor interface. The code is a C program for copying a file. It includes headers for stdio, fcntl, unistd, and stdlib. The main function opens 'source.txt' in read-only mode. It checks for errors. Then it opens 'destination.txt' in write-only, create, and truncate mode with a 0644 permission. It also checks for errors. A while loop reads from the source file into a 1024-byte buffer and writes it to the destination file. It checks for errors during writing. After the loop, it checks for errors during reading. Finally, it closes both files, prints a success message, and returns 0.

```
GNU nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source_fd, dest_fd;
    char buffer[1024];
    ssize_t bytes_read, bytes_written;

    source_fd = open("source.txt", O_RDONLY);

    if (source_fd == -1) {
        perror("Error opening source file");
        return 1;
    }

    dest_fd = open("destination.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (dest_fd == -1) {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0) {
        bytes_written = write(dest_fd, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read == -1) {
        perror("Error reading source file");
    }

    close(source_fd);
    close(dest_fd);

    printf("File copied successfully.\n");

    return 0;
}
```

```
saiviamsh@Varshini-PC: ~/p X + v
saiviamsh@Varshini-PC:~$ mkdir -p practical/week2/task1
saiviamsh@Varshini-PC:~$ cd practical/week2/task1
saiviamsh@Varshini-PC:~/practical/week2/task1$ pwd
/home/saiviamsh/practical/week2/task1
saiviamsh@Varshini-PC:~/practical/week2/task1$ ls
saiviamsh@Varshini-PC:~/practical/week2/task1$ nano pweek2.c
saiviamsh@Varshini-PC:~/practical/week2/task1$ echo "This is a sample file for demonstrating Linux system calls." > source.
txt
saiviamsh@Varshini-PC:~/practical/week2/task1$ cat source.txt
This is a sample file for demonstrating Linux system calls.
saiviamsh@Varshini-PC:~/practical/week2/task1$ gcc pweek2.c -o pweek2
saiviamsh@Varshini-PC:~/practical/week2/task1$ ls
pweek2 pweek2.c source.txt
saiviamsh@Varshini-PC:~/practical/week2/task1$ ./ pweek2
-bash: ./: Is a directory
saiviamsh@Varshini-PC:~/practical/week2/task1$ cat destination.txt
cat: destination.txt: No such file or directory
saiviamsh@Varshini-PC:~/practical/week2/task1$ ls -l
total 24
-rwxr-xr-x 1 saiviamsh saiviamsh 16224 Aug 25 21:21 pweek2
-rw-r--r-- 1 saiviamsh saiviamsh 1081 Aug 25 21:20 pweek2.c
-rw-r--r-- 1 saiviamsh saiviamsh 60 Aug 25 21:20 source.txt
saiviamsh@Varshini-PC:~/practical/week2/task1$ |
```

Observation: The program successfully copied the contents of source.txt to destination.txt using the open(), read(), write(), and close() system calls. The contents of both files were verified to be identical. During execution, the user-space program requested file operations from the Linux kernel through system calls, and the kernel performed the required file operations before returning control to the user-space program.

Task2: Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

[illegible]

```

rLimit6(0, RLIMIT_STACK, NULL, [rlim_cur=8192-1024, rlim_max=RLIM_INFINITY]) = 0
munmap(0x7f99a2090000, 9147) = 0
getrand(0x"v4u4a8xv1x0d0x31xblx31x53", 0, 0, GRND_NONBLOCK) = 8
brk(NULL) = 0
brk(0x7f99a2090000) = 0x7f99a2090000
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=2996, ...}) = 0
read(3, " = locale name alias data base.\n#...", 4096) = 2996
read(3, "", 4096) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=258, ...}) = 0
mmap(NULL, 258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2094000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27828, ...}) = 0
mmap(NULL, 27828, PROT_READ, MAP_SHARED, 3, 0) = 0x7f99a2086000
close(3) = 0
futex(0x7f99a2080472c, FUTEX_WAKE_PRIVATE, 2147483647) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=23, ...}) = 0
mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2093000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=47, ...}) = 0
mmap(NULL, 47, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2092000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=127, ...}) = 0
mmap(NULL, 127, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2091000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=62, ...}) = 0
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2090000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2085000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2084000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=270, ...}) = 0
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2083000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=1406, ...}) = 0
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2082000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2081000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2080000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/c-UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/c-utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=36000, ...}) = 0
mmap(NULL, 36000, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f99a2070000
close(3) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
open(AT_FDCWD, "sample.txt", O_RDONLY) = 54
fstat(3, {st_mode=S_IFREG|0644, st_size=54, ...}) = 0
fadvise64(3, 0, 0, POSIX_FADV_SEQUENTIAL) = 0
mmap(NULL, 139264, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f99a1dde000
read(3, "Hello, this is a sample file for...", 1072) = 54
read(3, "Hello, this is a sample file for...", 1072), this is a sample file for system call tracing.
) = 54
read(3, "", 131072) = 0
munmap(0x7f99a1dde000, 139264) = 0
close(3) = 0
close(1) = 0
close(2) = 0
exit_group(0) = ?
+++ exited with 0 +++
salvihaash@Varshini-PC:~/practical/week6/tasks2 |

```

```

saivihamsh@Varshini-PC:~/practical/week2/task2$ strace -o trace.txt cat sample.txt
Hello, this is a sample file for system call tracing.
saivihamsh@Varshini-PC:~/practical/week2/task2$ ls
sample.txt  trace.txt
saivihamsh@Varshini-PC:~/practical/week2/task2$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7fffb6451e08 /* 26 vars */) = 0
brk(NULL)                               = 0x64c4d0c61000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a1a98cab000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19147, ...}) = 0
mmap(NULL, 19147, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7a1a98ca6000
close(3)                                = 0

```

Observation: The strace utility successfully traced the system calls generated by the cat sample.txt command. The trace showed openat() being used to open sample.txt, followed by file and terminal operations using splice(), pipe(), and fcntl(). The close() system call was used to close the file descriptors, and exit_group(0) indicated successful program termination. The observed system calls demonstrate how the cat command interacts with Linux kernel services to access and display file contents.